Akamsha Reddy
DATA ENGINEER
Email: akamshareddy05@gmail.com |PH: 2142189259

PROFESSIONAL SUMMARY

- Almost 5 years in Data Engineering and implementing Hadoop, Spark, and cloud data warehousing solutions.
- Expertise in providing ETL solutions for diverse business models, ensuring data integration and transformation.
- Extensive experience with **Azure Data Factory**, **Azure Databricks**, **Azure Synapse Analytics**, and **Azure Storage** for building, deploying, and managing **end-to-end ETL/ELT pipelines**.
- Recreated application logic in Azure Data Lake, Data Factory, SQL Database, and SQL Data Warehouse for improved performance.
- Built **reporting data warehouses** and developed **scalable data pipelines** using **Azure Big Data**, **Hadoop Ecosystem**, and **Neo4j**.
- Integrated **Azure Event Grid** for real-time file indexing and event-based triggers, improving automation and reducing manual intervention.
- Developed and deployed **Azure Logic Apps** to automate alerts and stakeholder notifications, enhancing real-time data monitoring and communication.
- Built **Streamlit applications** for error tracking, leveraging **Azure Data Lake Storage (ADLS)** for centralized data management and reporting.
- Worked on **Azure CosmosDB** for NoSQL data management and integrated it seamlessly with **Azure Databricks** for advanced analytics.
- Developed **service principal scripts** using **Azure CLI** and **PowerShell** to ensure compliance and enhance security across **Azure applications**.
- Experience in **Azure DevOps** for CI/CD pipeline setup, infrastructure management using **Terraform**, and deployment automation.
- Utilized PL/SQL procedures in ETL transformations to optimize data processing for diverse business needs.
- Proficiency in AWS services such as EC2, S3, RDS, IAM, Glue, Redshift, DynamoDB, and more, enabling cloud-based data solutions.
- Experience with AWS databases, NoSQL databases, and data modeling, optimizing data storage and retrieval.
- Developed scalable systems using Hadoop technologies, including HDFS and MapReduce, for efficient data processing.
- Extensive experience with Informatica PowerCenter for data integration and transformation.
- Built database models, APIs, and views using Python to develop interactive web-based solutions.
- Designed and developed ETL processes using AWS Glue to migrate data from external sources into AWS Redshift.
- Leveraged AWS AppSync (GraphQL) for web API creation and data synchronization into Aurora Postgres or DynamoDB.
- Migrated ETL code from Talend to Informatica, ensuring a smooth transition of data processes.
- Backed up AWS Postgres to S3 on a daily basis using EMR, safeguarding data integrity.
- Tuned ETL jobs in the new environment, improving process efficiency after identifying bottlenecks.
- Designed Data Models and Dimensional Modeling for OLAP and Operational Data Store (ODS) applications.
- Tuned performance of Informatica mappings and sessions, optimizing data processing for better performance.
- Utilized SQL queries in BigQuery to extract valuable insights and perform data transformations as needed.
- Implemented software updates and patches for system maintenance, ensuring a healthy IT ecosystem.
- Strong understanding of Enterprise Data Warehouse best practices, ensuring data quality and consistency

- Built automation regression scripts for ETL process validation between multiple databases, enhancing data quality.
- Worked on migration projects that involved moving web methods code to the Informatica Cloud platform.
- Skilled in designing and implementing cost-effective and efficient ETL architectures.
- Conducted complex data analysis and provided critical reports to support various departments.
- Proficient in Business Intelligence tools like Business Objects and Data Visualization tools such as Tableau.

TECHNICAL SKILLS:

Programming/Scripting Languages	Scala, Java, Python, C, C++, SQL, Shell.
Big Data, Streaming & MQ	Hadoop, MapReduce, HDFS, Hive, Pig and Sqoop, Spark 2.0, Kinesis ,Data Streams, SQS, SNS.
AWS, Azure	EC2, S3, ECS, ECR, Lambda, CloudWatch, ELB, NLB, ALB, VPC, Route 53, IAM, Azure Data Factory, Databricks, Synapse, CosmosDB, Event Grid, Logic Apps, Data Lake Storage (ADLS), Azure SQL, Azure DevOps, Terraform, PowerShell, Key Vault, Monitor, Blob Storage, Functions, Stream Analytics, HDInsight, Azure CLI, Data Explorer
Other tools	Microsoft Office tools, VSTS, IoT, Git, Enterprise GitHub, Intellij, PyCharm, Maven.
Databases:	NoSQL, Oracle SQL, MSQL, RDBMS, Apache Cassandra, HBase, Snowflake
Big data Eco System	HDFS, Oozie, Zookeeper, Spark, Spark streaming, Kafka, NiFi
UNIX Tools:	Apache, Yum, RPM
FILE FORMATS	Txt, XML, JSON, Parquet, ORC, Protobuf
Cloud Computing	AWS, Azure
Visualization and Reporting Tools	Tableau, Microsoft Power BI, AWS Quick Sight.
Methodologies	Agile, UML, Design Patterns

PROFESSIONAL EXPERIENCE:

Microsoft – Redmond, WA
Data Engineer July 2024 to Present
Responsibilities:

- Spearheaded the Quantum Data Warehouse project, ensuring compliance with privacy standards (HIPAA, GDPR) by managing data flow diagrams, conducting privacy reviews, and integrating secure data handling practices.
- Developed and deployed end-to-end ETL pipelines using Azure Data Factory, Azure Databricks, and Synapse Analytics, optimizing data ingestion and transformation workflows from diverse data sources, including CosmosDB and Azure SQL.
- Created an Infrastructure Validation Page integrated with MS Lists to centralize error reporting and improve data validation processes, leading to faster issue identification and resolution.
- Implemented Logic Apps to automate alerts and notifications, facilitating real-time communication with stakeholders and reducing manual intervention in monitoring data pipeline health.
- Built dynamic Streamlit applications for error value visualization, enabling users to easily identify and resolve anomalies, significantly improving troubleshooting efficiency.

- Improved deployment processes by optimizing CI/CD pipelines with Azure DevOps and Terraform, leading to more reliable releases and reducing deployment times.
- Configured Synapse pipelines to automate data processing workflows, ensuring data consistency and improving data availability across reporting layers.
- Led the development of protocol bundles for efficient data organization and tracking, contributing to the DARPA preparation by reducing bundle processing times to approximately 1 second and ensuring system scalability to handle thousands of bundles per second.
- Enhanced file indexing performance by leveraging Event Grid for event-based triggers, improving real-time file tracking and reducing processing overhead.
- Collaborated with the security team to ensure compliance with internal policies related to App registrations, Service Principals, and data handling practices, including scripting for ownership management and property locking.
- Addressed complex data reconciliation challenges by developing robust validation frameworks, ensuring upstream and downstream data consistency across the data warehouse.
- Conducted performance tuning of Spark SQL queries and optimized resource allocation, resulting in a 30-40% increase in processing speed and improved system throughput.
- Provided technical documentation and training sessions for team members on new processes and tools, fostering a collaborative learning environment.
- Delivered presentations at internal meetings, such as the QHWD meeting, showcasing the infrastructure validation framework and its benefits to broader teams, fostering cross-team collaboration.

Macy's – Atlanta, GA **January 2022 to June 2023**
Data Engineer
Responsibilities

- Developed data pipelines for data gathering, transformation, and loading into data lakes or warehouses, a common practice in retail for data analysis.
- Created efficient Hive tables with static and dynamic partitions to manage and analyze retail data.
- Implemented ETL tasks to extract and load data from multiple sources, a fundamental process for retail data integration and analytics.
- Developed custom Kafka Connect connectors in Java or Python to integrate Kafka with proprietary or legacy systems, ensuring seamless data flow between disparate systems.
- Developed Spark applications for data extraction, transformation, and aggregation from various file formats commonly used in retail for big data processing.
- Worked on data visualization using Python and Tableau for presenting and interpreting retail data insights.
- Proficient in data ingestion techniques, including real-time and batch data ingestion, crucial for managing the constant flow of data.
- Implemented data caching strategies to reduce query response times, enabling faster and more responsive customer-facing applications, such as e-commerce websites and mobile apps.
- Implemented Snowflake data models for sales performance analytics, including sales trends analysis, product sales attribution, and geographic sales distribution analysis.
- Developed Map Reduce programs with Apache Hadoop for handling large volumes of data often found in industry.
- Managed Hadoop infrastructure with Cloudera Manager to ensure the stability and scalability of data processing in a retail environment.
- Created Python wrapper scripts for data extraction using Sqoop, a common practice in the retail sector for transferring data between Hadoop and relational databases.
- Created and maintained technical documentation for launching Hadoop clusters and executing Hive queries to ensure process reproducibility.

- Executed the essential task of extracting, transforming, and loading data from DB2 into a data warehouse using Teradata Client Utilities, a critical undertaking for effective retail data warehousing.
- Designed and implemented large-scale distributed solutions in AWS and GCP clouds, a common practice for retailers seeking scalability and cost-efficiency.
- Performed thorough impact assessments for system upgrades and migrations, strategically minimizing disruptions to the seamless flow of retail data processing workflows.
- Optimized SQL queries and data processing workflows to enhance the performance and efficiency of data operations.
- Worked on data modeling, including creating and maintaining data schemas, to support advanced analytics and reporting.
- Integrated Snowflake with BI tools like Tableau, Power BI, or Looker to build interactive dashboards and reports for retail sales performance, inventory management, and customer insights.

CloudTern Solutions – Hyderabad, India
Jr. Data Engineer **May 2019 to August 2021**
Responsibilities

- Built a **reporting data warehouse** from the ERP system, focusing on **Order Management** and **Invoice Service Contracts** modules to streamline product-related reporting and analytics.
- Developed **data models** and deployed **ETL pipelines** to meet CloudTern's business requirements, ensuring efficient data flow and integration across various systems.
- Utilized **Azure Big Data** and **Hadoop Ecosystem** components to construct **scalable data pipelines**, supporting CloudTern's data-driven solutions.
- Collaborated with reporting developers to implement **reporting solutions** and optimize **universe designs** for business intelligence platforms.
- Assisted **data scientists** and architects in building and maintaining data models using **Azure Databricks** for advanced data analysis and visualization.
- Developed data pipelines using tools like **Apache Flume**, **Sqoop**, and **Pig** to extract and store data from **web logs** and other sources, ensuring efficient data ingestion.
- Utilized **PowerShell** and **UNIX scripts** for automating file transfers, email communications, and other data management tasks to enhance operational efficiency.
- Participated in the **deployment process** from development to production environments, ensuring smooth and error-free rollouts of data solutions.
- Leveraged **Teradata SQL** and **Client Utilities** for efficient data migration and transformation into data warehouses, supporting business data needs.
- Integrated data from diverse sources, including **transactional databases**, **web logs**, and **third-party APIs**, providing comprehensive datasets for analysis.
- Collaborated with data scientists to develop **predictive models** for business insights, supporting strategic decision-making and resource optimization.
- Demonstrated expertise in **ETL processes** using **Apache Spark**, **Spark SQL**, and **Spark Streaming**, focusing on high-performance in-memory data processing.
- Developed customized **Airflow plugins** within Azure to enhance **workflow automation** and data pipeline management.
- Contributed to a **migration project** transitioning from **webMethods code** to **Informatica Cloud**, optimizing data integration workflows.

Education Details

- Completed master's in information systems & technology from the University of North Texas